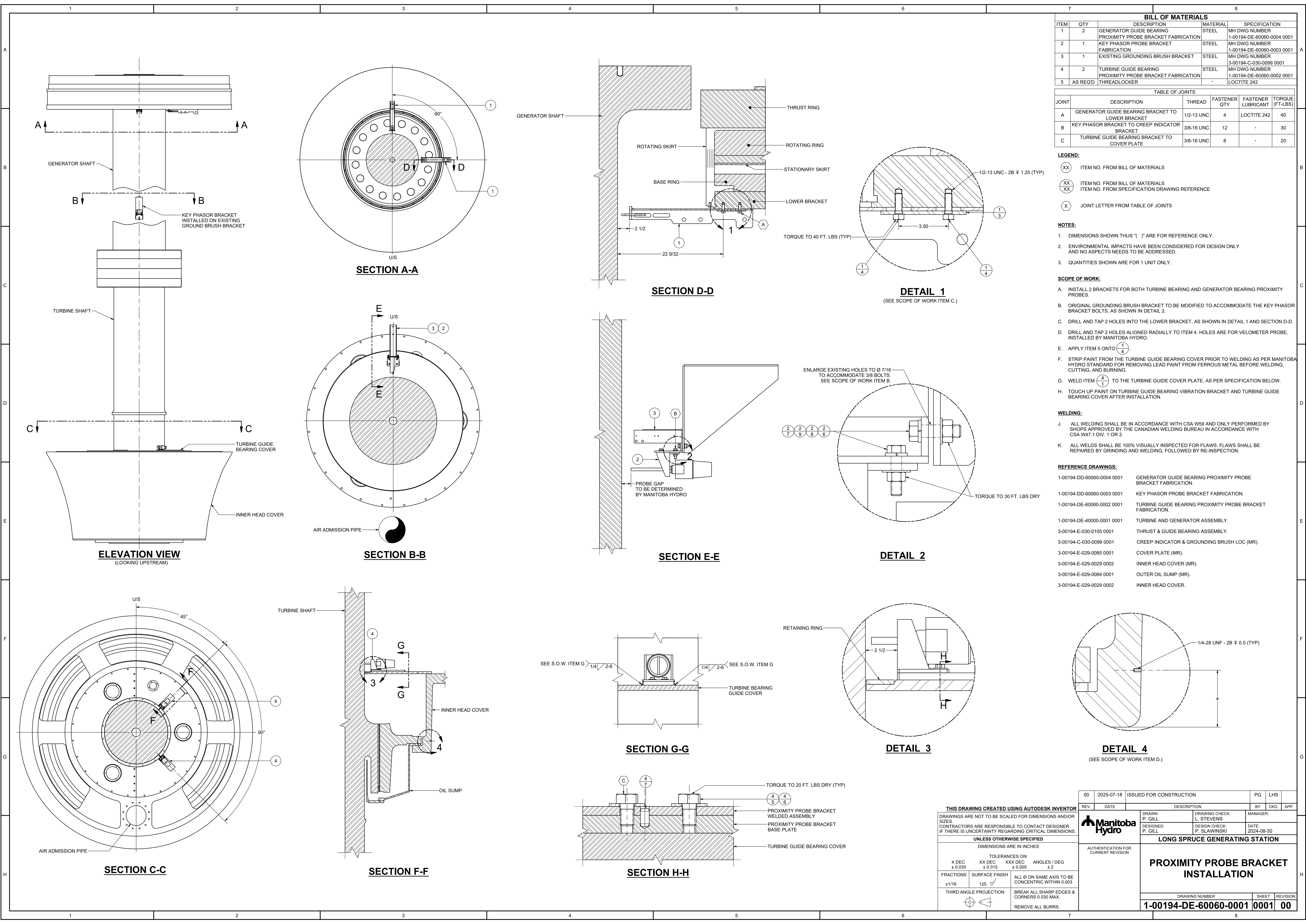




BILL OF MATERIALS				
ITEM	QTY	DESCRIPTION	MATERIAL	SPECIFICATION
1	2	GENERATOR GUIDE BEARING PROXIMITY PROBE BRACKET FABRICATION	STEEL	MH DWG NUMBER 1-00194-DE-60060-0004 0001
2	1	KEY PHASOR PROBE BRACKET FABRICATION	STEEL	MH DWG NUMBER 1-00194-DE-60060-0003 0001
3	1	EXISTING GROUNDING BRUSH BRACKET	STEEL	MH DWG NUMBER 3-00194-C-030-0099 0001
4	2	TURBINE GUIDE BEARING PROXIMITY PROBE BRACKET FABRICATION	STEEL	MH DWG NUMBER 1-00194-DE-60060-0002 0001
5	AS REQD	THREADLOCKER	-	LOCTITE 242

TABLE OF JOINTS					
JOINT	DESCRIPTION	THREAD	FASTENER QTY	FASTENER LUBRICANT	TORQUE (FT-LBS)
A	GENERATOR GUIDE BEARING BRACKET TO LOWER BRACKET	1/2-13 UNC	4	LOCTITE 242	40
B	KEY PHASOR BRACKET TO CREEP INDICATOR BRACKET	3/8-16 UNC	12	-	30
C	TURBINE GUIDE BEARING BRACKET TO COVER PLATE	3/8-16 UNC	8	-	20

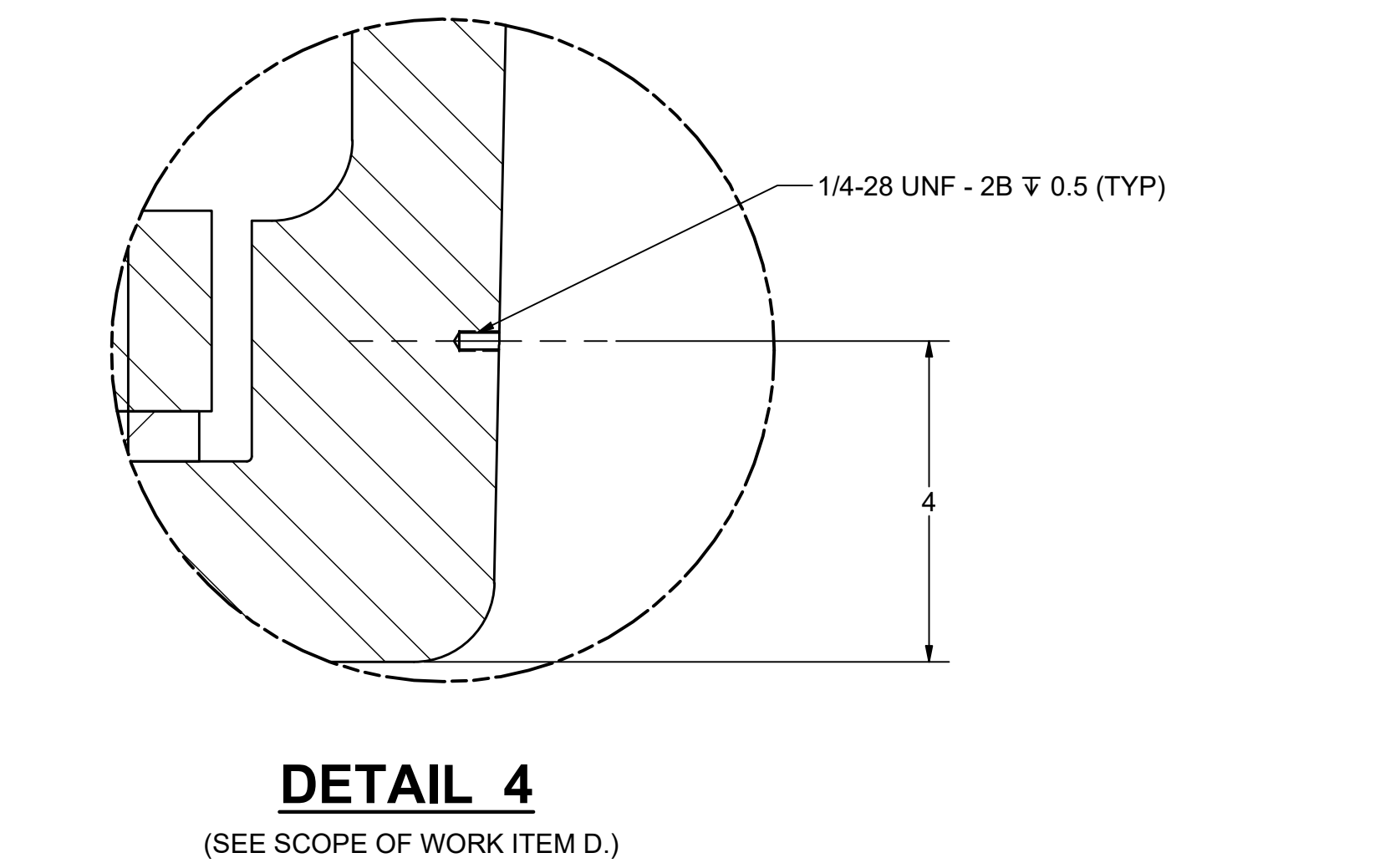
- LEGEND:**
- XX ITEM NO. FROM BILL OF MATERIALS
 - XX XX ITEM NO. FROM BILL OF MATERIALS
ITEM NO. FROM SPECIFICATION DRAWING REFERENCE
 - X JOINT LETTER FROM TABLE OF JOINTS

- NOTES:**
- DIMENSIONS SHOWN THUS "()" ARE FOR REFERENCE ONLY.
 - ENVIRONMENTAL IMPACTS HAVE BEEN CONSIDERED FOR DESIGN ONLY AND NO ASPECTS NEEDS TO BE ADDRESSED.
 - QUANTITIES SHOWN ARE FOR 1 UNIT ONLY.

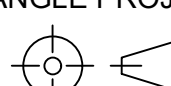
- SCOPE OF WORK:**
- INSTALL 2 BRACKETS FOR BOTH TURBINE BEARING AND GENERATOR BEARING PROXIMITY PROBES.
 - ORIGINAL GROUNDING BRUSH BRACKET TO BE MODIFIED TO ACCOMMODATE THE KEY PHASOR BRACKET BOLTS, AS SHOWN IN DETAIL 2.
 - DRILL AND TAP 2 HOLES INTO THE LOWER BRACKET, AS SHOWN IN DETAIL 1 AND SECTION D-D.
 - DRILL AND TAP 2 HOLES ALIGNED RADIALLY TO ITEM 4. HOLES ARE FOR VELOMETER PROBE, INSTALLED BY MANITOBA HYDRO.
 - APPLY ITEM 5 ONTO (1/4).
 - STRIP PAINT FROM THE TURBINE GUIDE BEARING COVER PRIOR TO WELDING AS PER MANITOBA HYDRO STANDARD FOR REMOVING LEAD PAINT FROM FERROUS METAL BEFORE WELDING, CUTTING, AND BURNING.
 - WELD ITEM (4/7) TO THE TURBINE GUIDE COVER PLATE, AS PER SPECIFICATION BELOW.
 - TOUCH UP PAINT ON TURBINE GUIDE BEARING VIBRATION BRACKET AND TURBINE GUIDE BEARING COVER AFTER INSTALLATION.

- WELDING:**
- ALL WELDING SHALL BE IN ACCORDANCE WITH CSA W59 AND ONLY PERFORMED BY SHOPS APPROVED BY THE CANADIAN WELDING BUREAU IN ACCORDANCE WITH CSA W47.1 DIV. 1 OR 2.
 - ALL WELDS SHALL BE 100% VISUALLY INSPECTED FOR FLAWS. FLAWS SHALL BE REPAIRED BY GRINDING AND WELDING, FOLLOWED BY RE-INSPECTION.

- REFERENCE DRAWINGS:**
- | | |
|----------------------------|--|
| 1-00194-DD-60060-0004 0001 | GENERATOR GUIDE BEARING PROXIMITY PROBE BRACKET FABRICATION. |
| 1-00194-DD-60060-0003 0001 | KEY PHASOR PROBE BRACKET FABRICATION. |
| 1-00194-DE-60060-0002 0001 | TURBINE GUIDE BEARING PROXIMITY PROBE BRACKET FABRICATION. |
| 1-00194-DE-40000-0001 0001 | TURBINE AND GENERATOR ASSEMBLY. |
| 3-00194-E-030-0105 0001 | THRUST & GUIDE BEARING ASSEMBLY. |
| 3-00194-C-030-0099 0001 | CREEP INDICATOR & GROUNDING BRUSH LOC (MR). |
| 3-00194-E-029-0085 0001 | COVER PLATE (MR). |
| 3-00194-E-029-0029 0002 | INNER HEAD COVER (MR). |
| 3-00194-E-029-0084 0001 | OUTER OIL SUMP (MR). |
| 3-00194-E-029-0029 0002 | INNER HEAD COVER. |



00	2025-07-18	ISSUED FOR CONSTRUCTION	PG	LHS	
REV.	DATE	DESCRIPTION	BY	CKD.	APP.
		DRAWN: P. GILL DESIGNED: P. GILL	DRAWING CHECK: L. STEVENS DESIGN CHECK: P. SLAWINSKI	MANAGER:	DATE: 2024-08-30
LONG SPRUCE GENERATING STATION					
PROXIMITY PROBE BRACKET INSTALLATION					
DRAWING NUMBER				SHEET	REVISION
1-00194-DE-60060-0001				0001	00

THIS DRAWING CREATED USING AUTODESK INVENTOR					
DRAWINGS ARE NOT TO BE SCALED FOR DIMENSIONS AND/OR SIZES. CONTRACTORS ARE RESPONSIBLE TO CONTACT DESIGNER IF THERE IS UNCERTAINTY REGARDING CRITICAL DIMENSIONS.					
UNLESS OTHERWISE SPECIFIED					
DIMENSIONS ARE IN INCHES					
TOLERANCES ON:					
X DEC ± 0.030		XX DEC ± 0.015		XXX DEC ± 0.005	
ANGLES / DEG ± 2					
FRACTIONS ±1/16		SURFACE FINISH 125 ✓		ALL Ø ON SAME AXIS TO BE CONCENTRIC WITHIN 0.003	
THIRD ANGLE PROJECTION		BREAK ALL SHARP EDGES & CORNERS 0.030 MAX.			
		REMOVE ALL BURRS.			